FinTech Payment Failure & Transaction Decline Analytics

📌 Project Overview

Payment failures and transaction declines can directly impact customer experience, merchant revenue, and payment platform performance.

This project analyzes payment transaction data to identify the key factors contributing to transaction failures and declines. The analysis focuses on transaction success rates, failure patterns, payment methods, banks, failure reasons, cities, devices, merchants, and monthly trends.

The project uses **Python for data generation and preparation, SQL for data analysis, and Google Sheets for KPI reporting and dashboard visualization.**

🎯 Business Problem

A FinTech payment platform wants to understand:

- What percentage of transactions are successful?
- What is the overall transaction failure rate?
- Which payment methods have the highest failure rates?
- Which banks are experiencing the most transaction failures?
- What are the most common failure reasons?
- Which cities, devices, and merchants have higher failure rates?
- How does the failure rate change over time?
- How much transaction value is associated with failed transactions?
- Where should the business focus its efforts to improve payment success?

🎯 Project Objectives

The main objectives of this project are to:

1. Measure overall payment transaction performance.
2. Calculate transaction success and failure rates.
3. Identify high-risk payment methods and banks.
4. Analyze the major reasons for transaction failures.
5. Identify monthly trends in transaction failures.
6. Analyze failure patterns across cities, devices, merchants, and card types.
7. Quantify the monetary value associated with failed transactions.
8. Build a management-friendly KPI dashboard.
9. Generate actionable business recommendations.

🛠️ Tools & Technologies

Tool	Purpose
Python	Data generation, cleaning and validation
Pandas	Data manipulation
NumPy	Synthetic data generation
SQL	Data analysis and KPI calculations
SQLite	SQL analysis environment
Google Sheets	Dashboard and visualization
GitHub	Project documentation and version control

📊 Dataset

The project uses a synthetic payment transaction dataset containing transaction-level information.

Key Columns

Column	Description
transaction_id	Unique transaction identifier
customer_id	Unique customer identifier
merchant_id	Merchant identifier
transaction_date	Date and time of transaction
amount	Transaction amount
payment_method	UPI, Credit Card, Debit Card, etc.
bank	Customer/payment processing bank
card_type	Visa, Mastercard, RuPay, etc.
device	Mobile, Desktop or Tablet
city	Customer transaction city
status	SUCCESS, FAILED, DECLINED or TIMEOUT
failure_reason	Reason for unsuccessful transaction
processing_time	Transaction processing time

📈 Key KPIs

The dashboard tracks the following key payment performance indicators:

- **Total Transactions**
- **Successful Transactions**
- **Failed Transactions**
- **Success Rate**
- **Failure Rate**
- **Total Transaction Value**
- **Failed Transaction Value**
- **Average Transaction Value (AOV)**

🔍 Analysis Performed

1. Transaction Status Analysis

Analyzed the distribution of:

- Successful transactions
- Failed transactions
- Declined transactions
- Timeout transactions

2. Payment Method Analysis

Compared transaction performance across:

- UPI
- Credit Card
- Debit Card
- Net Banking
- Wallet

Metrics analyzed include transaction volume, failed transactions and failure rate.

3. Bank Analysis

Analyzed bank-wise:

- Transaction volume
- Failed transactions
- Failure rate
- Bank ranking based on failure rate

4. Failure Reason Analysis

Identified the most common reasons behind unsuccessful transactions, including:

- Insufficient Funds
- Bank Server Down
- Technical Error
- Invalid Card
- Fraud Suspected
- Transaction Limit Exceeded
- Network Timeout

5. Monthly Trend Analysis

Analyzed monthly:

- Transaction volume
- Failed transaction volume
- Failure rate

This helps identify whether payment performance is improving or deteriorating over time.

6. City & Device Analysis

Compared transaction failures across different:

- Cities
- Devices
- Card types

7. High-Value Failure Analysis

Identified high-value failed transactions to understand the potential monetary impact of payment failures.

💡 Business Insights

The analysis can help a payment platform identify:

- Payment methods with unusually high failure rates.
- Banks contributing disproportionately to failed transactions.
- The most common technical and operational failure reasons.
- Periods where transaction failures increase significantly.
- High-value transactions affected by payment failures.
- Devices, cities or merchants showing unusual failure patterns.

These findings can be used to prioritize payment infrastructure improvements, bank integrations, fraud-rule reviews, and customer experience initiatives.

🚀 Business Recommendations

Based on the analysis framework, potential recommendations include:

1. Improve High-Failure Bank Integrations

Investigate banks with consistently high failure rates and review API/integration reliability.

2. Monitor Payment Methods

Set up monitoring for payment methods whose failure rates exceed the acceptable threshold.

3. Reduce Technical Failures

Investigate technical errors, network timeouts and bank-server issues to improve payment reliability.

4. Monitor High-Value Failures

Prioritize investigation of high-value failed transactions because they can have a greater financial impact.

5. Build Automated Monitoring

Create automated alerts when payment failure rates cross predefined thresholds.

📂 Project Structure

```
fintech-payment-failure-analytics/
│
├── data/
│   └── transactions.csv
│
├── sql/
│   ├── transaction_summary.sql
│   ├── success_failure_analysis.sql
│   ├── payment_method_analysis.sql
│   ├── bank_analysis.sql
│   ├── failure_reason_analysis.sql
│   ├── monthly_analysis.sql
│   └── high_value_failures.sql
│
├── python/
│   └── data_generation_and_cleaning.ipynb
│
├── dashboard/
│   └── payment_failure_dashboard.xlsx
│
├── screenshots/
│   └── dashboard.png
│
├── insights/
│   └── business_insights.md
│
└── README.md
```

🔄 Project Workflow

Raw Transaction Data
↓
Python Data Generation & Cleaning
↓
SQLite Database
↓
SQL Analysis
↓
KPI & Business Metrics
↓
Google Sheets Dashboard
↓
Business Insights & Recommendations

📌 Key Learning Outcomes

Through this project, I practiced:

- SQL aggregation and conditional logic
- GROUP BY and HAVING
- CASE WHEN
- CTEs
- Window functions
- Ranking
- Date/time analysis
- KPI calculation
- Data validation
- Python/Pandas data manipulation
- Dashboard development
- Business problem solving
- Translating data into actionable insights

👤 Author

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⭐ Project Goal

The goal of this project is to demonstrate how transaction-level payment data can be transformed into **actionable business insights using SQL, Python and dashboarding tools.**